

Mechanism-Constrained Inference of Voting Fairness in Competitive Entertainment Systems

Summary

Dancing with the Stars (DWTS), as a globally renowned program, employs a scoring mechanism that combines professional judges' opinions with public votes. However, the alternating use of “ranking summation” and “percentage summation” systems across different seasons often leads to controversial outcomes. This study conducts an in-depth analysis by constructing a multi-stage mathematical model and proposes an optimized scoring system.

For Task 1, we develop a **constrained Bayesian inverse model** that integrates a Dirichlet prior with Markov Chain Monte Carlo (MCMC) sampling to estimate the weekly fan voting proportions of contestants across 34 seasons. Model validation indicates an overall consistency rate of **91.38%**, demonstrating that the proposed approach can accurately reproduce historical competition outcomes and thus provides a reliable data foundation for subsequent analyses.

For Task 2, we develop **contestant-level survival trajectories** and apply a **combination of the Cox proportional hazards model** and a **panel fixed-effects model**. The results indicate that the percentage-based scoring mechanism provides a significant protective effect for highly popular contestants, increasing their average survival duration by 0.36 weeks compared with the rank-based system. Moreover, under the percentage-based mechanism, a 0.1 increase in fan support reduces the elimination risk by 32.9%, substantially higher than the 17.8% observed under the rank-based mechanism. These findings suggest that the percentage-based system places greater emphasis on fan voting influence.

For Task 3, simulation analyses of controversial contestants such as Jerry Rice and Bobby Bones demonstrate that the method used to combine judges' scores and audience votes directly affects advancement outcomes. Incorporating a mechanism in which judges select the eliminated contestant from the bottom two further reduces the elimination risk of highly popular contestants, leading to results that are more professionally grounded and persuasive.

For Task 4, we develop a **SHAP-enhanced XGBoost model** to conduct decision-level analysis, which reveals systematic preference differences between judges and fans (Fig .11). The results indicate that judges place greater emphasis on contestant age and performance maturity, whereas fans tend to prioritize partner coordination and on-stage chemistry.

For Task 5, we propose a **Dynamic Weighted Scoring System (DWSS)** that integrates adaptive weight adjustment, progress-based bonuses, and a “danger zone” rescue mechanism. Validation using historical data shows that the system correctly places **87.85%** of actually eliminated contestants within the danger zone, substantially mitigating the influence of extreme voting behaviors. These results demonstrate the reliability and feasibility of the proposed scoring framework, while effectively balancing competitive professionalism and entertainment value.

This study not only provides a scientifically grounded framework for optimizing the scoring rules of DWTS, but also offers a novel methodological perspective for the integration of multi-source evaluation information.

Keywords: Constrained Bayesian Inversion Model; Cox Proportional Hazards Model; SHAP-enhanced XGBoost Model; Dynamic Weighted Scoring System (DWSS)

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1 Introduction

1.1 Problem Background

Dancing with the Stars (DWTS) is one of the most successful reality television franchises worldwide. Its core mechanism combines evaluations from professional judges with large-scale audience voting to determine contestant eliminations and final rankings. Across multiple seasons of the U.S. edition, different aggregation rules have been employed to merge judges' scores and fan votes, primarily through rank-based and percentage-based systems. These rule changes have repeatedly led to controversial outcomes in which professional assessments diverge sharply from audience preferences. At a deeper level, this phenomenon reflects a fundamental challenge in multi-source evaluation fusion: how to establish a fair and reasonable balance between expert judgment and public opinion. By leveraging historical competition data and constructing mathematical models, it becomes possible to infer latent audience voting distributions, systematically assess the fairness of alternative scoring mechanisms, and provide a theoretical foundation for designing scoring systems that are both more rational and more interpretable.



Figure 1: DWTS Art Direction Nomination

1.2 Clarifications and Restatements

Given the background information and constraints of the problem, we must complete the following tasks:

Task 1: Develop a mathematical model to estimate the weekly fan voting proportions of contestants across seasons, and provide a consistency metric to evaluate model performance, together with a quantitative assessment of the uncertainty in the estimated results.

Task 2: Using the estimated fan voting data, compare the performance of the rank-sum and percentage-sum aggregation methods across all seasons, and analyze which approach places greater emphasis on fan voting.

Task 3: For representative controversial contestants, examine whether outcomes change under alternative scoring rules or when incorporating a mechanism in which judges select the eliminated contestant from the bottom two, and provide recommendations for an improved scoring policy.

Task 4: Develop a model to analyze the effects of estimated fan votes, judges' scores, and contestant characteristics on performance outcomes, and examine whether their influences on judges' evaluations and fan voting are consistent.

Task 5: Propose a more fair and robust system for combining judges' scores with fan voting and determining eliminations, and demonstrate its advantages using historical data.

1.3 Our Work

The study first constructs a constrained Bayesian inverse model, using MCMC rejection sampling to obtain posterior distributions, enabling high-precision estimation of previously unknown fan voting proportions. Next, based on the estimated fan votes, contestant-level survival trajectories are built, and the Cox proportional hazards model is employed to quantify the effect of fan support on elimination risk, identify which scoring mechanism provides stronger protection for highly popular contestants, and evaluate its impact on controversial participants. Subsequently, an XGBoost model enhanced with SHAP is developed to analyze how estimated fan votes, judges' scores, and contestant characteristics influence performance outcomes. Finally, a Dynamic Weighted Scoring System (DWSS) is proposed. For clarity and visualization, Fig. 2 illustrates the overall research framework.

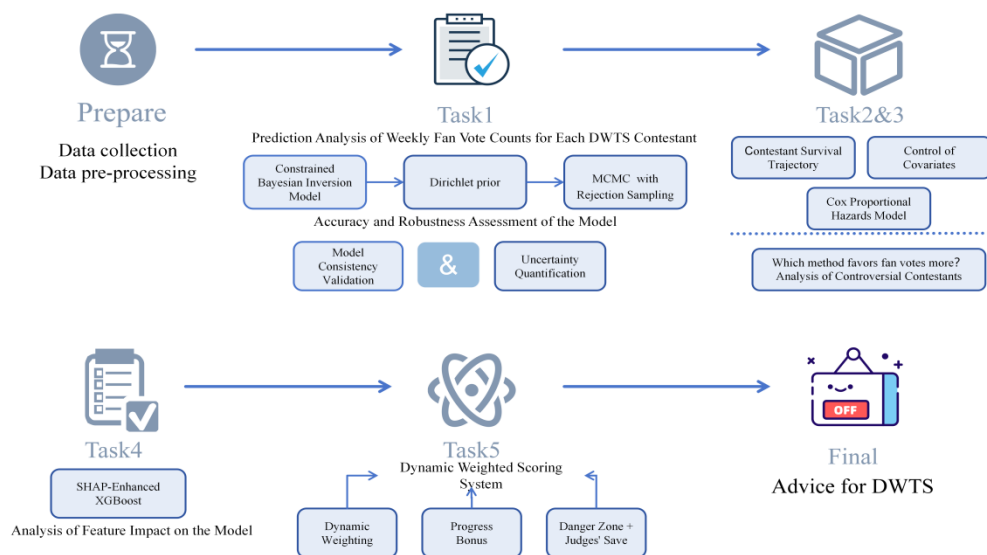


Figure 2: The Flow Chart of Our Work

2 Preparation for Modeling

2.1 Model Assumptions

Some fundamental assumptions are listed below.

- **Assumption 1:** Audience voting is positively correlated, to some extent, with judges' scores.

Justification: High-quality performances by contestants are generally recognized by both professional judges and the general audience.

- **Assumption 2:** Judges' scores reasonably reflect the contestants' dance performance in a given week.

Justification: Although judges' evaluations involve some subjectivity, they provide an effective approximation of dance quality and do not exhibit systematic bias toward any particular

type of contestant.

- **Assumption 3:** Contestant eliminations strictly follow the officially announced rules.

Justification: Weekly elimination outcomes are determined solely by the combination of judges' scores and fan votes according to the prescribed rules, with no human intervention or random adjustments involved.

- **Assumption 4:** All contestants compete under the same conditions, and no external factors.

Justification: As a professionally produced television show, DWTS provides standardized venues, lighting, sound, and other facilities.

2.2 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
s	Season index
t	Week index
i	Contestant index
k	Number of Active Contestants in the Week
α	Dirichlet prior concentration vector
Ω	Feasible solution space
D	Observed data set (including judges' scores, elimination outcomes, final rankings, etc.)
ϕ_j	SHAP value of a feature j

2.3 Data Preprocessing

To ensure the accuracy and reliability of model solutions, data preprocessing is required to provide high-quality input for subsequent model development. The detailed steps are as follows:

- (1) **Handling Missing Values:** For judges who did not provide scores in certain weeks, "N/A" entries are replaced with 0 rather than removing the contestant or week, in order to preserve the completeness of the season.
- (2) **Consistency Handling for Elimination Indicators and Time Series:** Contestants who receive a score of 0 in subsequent weeks are marked as "eliminated". During modeling, only data prior to elimination are used, preventing later invalid records from affecting the model.
- (3) **Standardization of Special Scoring Mechanisms:** For special cases such as team dances, bonus points (e.g., dance-offs), and multiple dance routines, scores are standardized to eliminate the effects of differing scales.
- (4) **Encoding of Categorical Variables:** Categorical variables such as professional category (athlete, actor, singer, etc.), nationality (U.S./other countries), and partner ID are converted into numerical form. Label encoding is applied to map string features to numeric values, facilitating subsequent modeling.

3 Task 1: Constrained Bayesian Inversion Model

This study aims to estimate the weekly fan vote share for each contestant in the 34 seasons of *Dancing with the Stars* (DWTS). By characterizing the multivariate discrete distribution of fan vote proportions and leveraging the constraints imposed by competition outcomes, we adopt an innovative Bayesian^{3,4} framework combined with Markov Chain Monte Carlo (MCMC) sampling. A Dirichlet prior distribution is incorporated to enhance the model, with the goal of constructing a high-precision estimator for fan voting while quantifying the consistency and certainty of the estimates. Utilizing publicly available data on judges' scores, elimination results, and final rankings across all 34 seasons, we obtain the posterior distributions and point estimates of weekly fan vote proportions for each contestant. These results provide a robust data foundation for subsequent analyses of voting mechanisms and the exploration of influencing factors.

3.1 Model Constrution

1. Dirichlet-Based Modeling

Suppose there are k active contestants in a given week, and their fan voting proportion vector is:

$$V = (V_1, \text{dots}, V_k), V_i \geq 0, \sum_i V_i = 1 \quad (1)$$

$$V \sim \text{Dirichlet}(\alpha) \quad (2)$$

where V_i represents the latent proportion of fan votes for the i -th contestant.

The Dirichlet distribution is defined on the probability simplex, naturally satisfying the non-negativity and sum-to-one constraints of proportion vectors. It also forms a conjugate structure with the multinomial distribution, making it well-suited for Bayesian modeling of voting proportions.

The concentration parameter α of the Dirichlet distribution determines the shape of the prior. To avoid subjective parameter settings, this study fits α in a data-driven manner based on historical judge score proportions.

For historical week t , define the judge score proportion of the i -th contestant as:

$$S_{i,t} = \frac{J_{i,t}}{\sum_j J_{j,t}} \quad (3)$$

where $J_{i,t}$ denotes the total judge score for that week.

Using the expectation property of the Dirichlet distribution, Construct a least squares objective function.

$$\mathbb{E}[V_i] = \frac{\alpha_i}{\sum_j \alpha_j} \quad (4)$$

$$\min_{\alpha} \sum_{t=1}^T \left| \mathbf{s}_t - \frac{\alpha}{\sum_j \alpha_j} \right|^2 \quad (5)$$

2. Total Score Calculation Model

In the 34 seasons of *Dancing with the Stars* (DWTS), the rules for combining judges' scores and fan votes have evolved into three types: the percentage system, the ranking system, and the ranking system with a save mechanism. Each system features distinct logic for calculating overall scores and determining eliminations. In this section, we construct an overall score calculation model fully aligned with the season-specific rules, formalizing the core logic of all three systems through mathematical expressions. Implementation details are verified against actual competition procedures to ensure consistency, providing a standardized computational framework for subsequent fan vote estimation and comparative analysis of rule effects.

- Percentage System

The percentage system integrates judges' scores and fan votes through an "absolute proportion summation," calculating the overall score as:

$$C_i = \frac{J_i}{\sum_j J_j} + V_i \quad (6)$$

- Ranking System

The ranking system integrates judges' scores and fan votes through a "relative ranking summation":

$$C_i = R(J_i) + R(V_i) \quad (7)$$

Here, $R(\cdot)$ denotes the dense ranking function.

- Judge Save Mechanism

This system is an extension of the ranking system, with the core difference being that the elimination rule changes from "directly eliminating the lowest-scoring contestant" to "first selecting the bottom two contestants, then letting the judges decide who is eliminated":

$$i \in \operatorname{argmax}_2(C_i) \quad (8)$$

3. Posterior Distribution and Bayesian Estimation

Let the observed data be $\mathcal{D}$, which includes elimination results and final rankings. The posterior distribution is then:

$$p(V | \mathcal{D}) \propto p(\mathcal{D} | V) p(V) \quad (9)$$

Define the deterministic likelihood function:

$$p(\mathcal{D} | V) = \begin{cases} 1 & \text{if the model result matches the true result,} \\ 0 & \text{otherwise.} \end{cases} \quad (10)$$

Therefore, the posterior distribution can be expressed as:

$$p(V | \mathcal{D}) \propto \operatorname{Dirichlet}(\alpha) \cdot \mathbb{I}(V \in \Omega) \quad (11)$$

Here, Ω denotes the feasible solution space induced by the competition rules. The posterior is essentially a truncated Dirichlet distribution constrained by the mechanism.

Obtain posterior samples via MCMC and compute the posterior mean and standard deviation:

$$\hat{V}_i = \frac{1}{m} \sum_{j=1}^m V_i^{(j)}, \quad \sigma_i = \sqrt{\frac{1}{m} \sum_{j=1}^m (V_i^{(j)} - \hat{V}_i)^2} \quad (12)$$

4. Design of Evaluation Metrics

To evaluate the reliability and validity of the model's inference results, this study constructs an assessment framework from two perspectives: uncertainty quantification and consistency verification.

- Posterior Uncertainty Quantification

Based on the posterior distribution obtained via MCMC sampling, the 95% posterior credible interval for the fan vote proportion $V_{i,s,t}$ of each contestant in week t is calculated as:

$$CI_{95\%}(V_{i,s,t}) = [q_{0.025}(V_{i,s,t} | Data), q_{0.975}(V_{i,s,t} | Data)] \quad (13)$$

Here, $q_p(\cdot)$ denotes the p -th quantile of the posterior distribution, and $Data$ represents the set of observed data, including judges' scores, elimination results, and competition rules.

This interval reflects the model's degree of uncertainty regarding the true fan vote proportion under the imposed constraints; a narrower interval indicates more stable and reliable inference results.

- Elimination Outcome Consistency Metric

To evaluate the model's explanatory power under the actual competition rules, we define the elimination prediction accuracy metric:

$$Accuracy = \frac{\sum_{s,t} \mathbb{I}(\hat{E}_{s,t} = E_{s,t})}{\sum_{s,t} 1} \quad (14)$$

Here, $E_{s,t}$ denotes the set of eliminated contestants in season s and week t ; $\hat{E}_{s,t}$ represents the elimination results inferred from the posterior mean of votes combined with the competition rules; $\mathbb{I}(\cdot)$ is an indicator function that takes the value 1 if the prediction matches the actual result and 0 otherwise.

This metric measures the overall consistency between the model's inferred outcomes and the actual competition results at the mechanism level and serves as a key standard for evaluating the model's validity and structural effectiveness.

3.2 Model Solution

Using the Bayesian framework combined with MCMC sampling, we estimated the weekly fan vote proportions for each season. Taking Week 3 of Season 2 as an example, the estimated fan vote proportions for the 8 active contestants are presented in the Table2:

From the results of this week, it can be seen that Jerry Rice ranked second in terms of estimated fan vote proportion. Although his judges' score was at a mid-level, the high fan vote proportion compensated for the technical score disadvantage, helping him advance steadily and ultimately finish as the runner-up in Season 2, fully consistent with the actual competition results.

Table 2: predicted results

Contestant	Estimated vote proportion	Standard deviation ($\pm$)
Tia Carrere	0.1243	0.0124
George Hamilton	0.1266	0.0118
Lisa Rinna	0.1236	0.0105
Stacy Keibler	0.1261	0.0133
Jerry Rice	0.1330	0.0104
Giselle Fernandez	0.1062	0.0072
Master P	0.1342	0.0080
Drew Lachey	0.1259	0.0129

From the results of this week, it can be seen that Jerry Rice ranked second in terms of estimated fan vote proportion. Although his judges' score was at a mid-level, the high fan vote proportion compensated for the technical score disadvantage, helping him advance steadily and ultimately finish as the runner-up in Season 2, fully consistent with the actual competition results.

Meanwhile, the results also reflect the uncertainty captured by the model: Giselle Fernandez exhibits the lowest standard deviation for this week, indicating high certainty in her estimated vote proportion, whereas Stacy Keibler shows a relatively larger standard deviation, reflecting a wider range of fluctuation in her fan votes. Nevertheless, the estimates remain within a reasonable range, not affecting the overall assessment of her popularity.

Visualizing the results further validates the model's ability to fit scenarios with disparities between "popularity and technical skill," demonstrating strong alignment with the actual competition progression.

As shown in the Figs. 3(a), the chart illustrates the trend of Jerry Rice's fan vote proportion over the weeks in Season 2 (red curve). Although Jerry Rice's judges' scores were at mid-level for most weeks, the model-predicted fan vote proportion exhibited a continuous upward trend, rising from approximately 0.1 in Week 1 to over 0.35 by Week 8, consistently surpassing other contestants in the same season. This result aligns with the actual competition outcome: Jerry Rice leveraged his high popularity to offset technical score shortcomings, ultimately advancing to the finals and securing the runner-up position.

As shown in the Figs. 3(b), the chart depicts Bobby Bones' judges' scores (blue curve, left axis) and fan vote proportions (red curve, right axis) in Season 27. Bobby's judges' scores remained consistently low throughout the season (below 25 points in most weeks), yet the model-predicted fan vote proportion stayed higher than that of other contestants, with a notable increase in later weeks. This result accurately reproduces the actual competition outcome—Bobby Bones leveraged overwhelming fan support to overcome low judges' scores and ultimately win the championship—validating the model's effectiveness in capturing "popularity-driven advancement" scenarios.

3.3 Model Evaluation:

Based on prior accuracy and posterior uncertainty metrics, and combined with season-long visualizations, the overall model performance was evaluated from two dimensions: consistency and certainty.

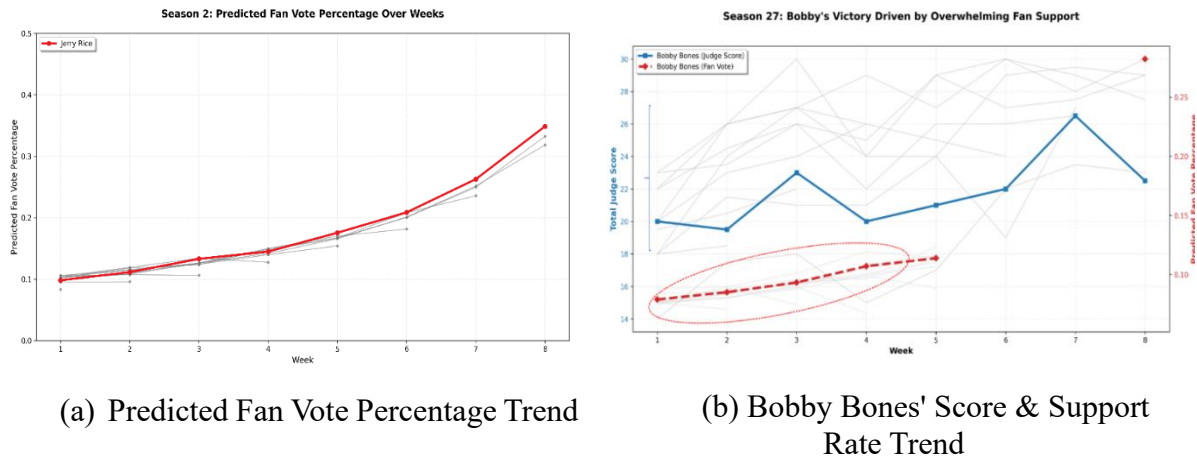


Figure 3: Contestant Fan Support & Score Trend Comparison

Based on prior accuracy and posterior uncertainty metrics, and combined with season-long visualizations, the overall model performance was evaluated from two dimensions: consistency and certainty.

As shown in the Figs. 4(a), the average model consistency across all seasons is 91.38%. For seasons using the ranking-based aggregation method (Seasons 1–2 and 28–34), the average consistency is 92.06%, while for seasons using the percentage-based aggregation method (Seasons 3–27), it is 90.7%. Weeks without valid samples account for only 1.8%, indicating that the model estimates can accurately reproduce the actual competition progression. This fully demonstrates the model's adaptability to different seasonal scoring mechanisms and its reliability in replicating real competition outcomes.

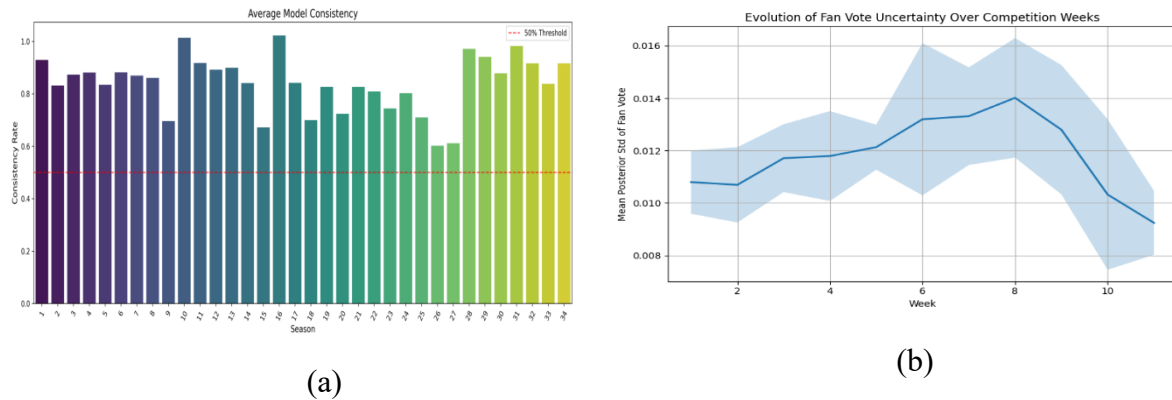


Figure 4: Model Consistency & Fan Vote Uncertainty Trends

As shown in the Figs. 4(b), the posterior standard deviation of the estimated fan vote proportions is plotted across competition weeks, with the blue curve representing the weekly average posterior standard deviation and the light blue shaded area indicating the range of variation for each week. Specifically, in the early stage of the competition (Weeks 1–4), the posterior standard deviation gradually increases due to the larger number of active contestants and relatively weaker observational constraints, resulting in higher estimation dispersion. During the mid-stage (Weeks 5–8), the posterior standard deviation peaks and then begins to decline. As contestants are progressively eliminated, the observational constraints on the remaining contestants strengthen, and the sample concentration of the estimates increases significantly. In the

late stage (Weeks 9–10), the posterior standard deviation drops sharply, reaching its lowest point during the finals week, indicating that the strong observational constraints imposed by the final rankings most effectively refine the prior distribution. At this stage, the certainty of the fan vote estimates is at its highest.

4 Task 2&3: Effect of Scoring Systems on Contestant Progression

This study focuses on the 34 seasons of DWTS to compare the effects of percentage-based versus rank-based scoring mechanisms on contestants' advancement trajectories. Using publicly available data on judges' scores, fan support, and elimination outcomes, contestant-level survival trajectories were constructed, and both the Cox proportional hazards model and panel fixed-effects model were employed to quantify the impact of fan support on elimination risk and survival weeks. The Cox model does not require a pre-specified baseline hazard, providing flexibility and allowing estimation of the marginal effect of fan support on elimination risk, while the panel fixed-effects model controls for systematic differences across seasons, enhancing the robustness of mechanism comparisons. However, these methods are limited by the proportional hazards assumption, which may overlook dynamic variations in mechanism effects. To address this limitation, the study further incorporates quantile group comparisons and marginal effect analyses, and subsequently introduces SHAP-based interpretable machine learning to nonlinearly attribute high-dimensional contestant features, thereby preserving the interpretability of statistical models while better capturing complex mechanism structures.

4.1 Construction of Contestant-Level Trajectories

To quantify the impact of the two scoring mechanisms on contestant advancement, we first constructed survival trajectories for each contestant, recording their weekly participation status and elimination outcomes under the different mechanisms.

For any contestant C_i their survival trajectory is defined as the set of weekly status indicators:

$$T_i = \{(w, J_{i,w}, F_i, \delta_{i,w}^{\text{percent}}, \delta_{i,w}^{\text{rank}}) \mid w = 1, 2, \dots, W_i\} \quad (15)$$

Here, W_i denotes the actual number of weeks contestant C_i participated; $J_{i,w}$ represents the total judges' score for contestant C_i in week w ; F_i is the fan support level for contestant C_i ; and $\delta_{i,w}^{\text{rank}}$ is the elimination indicator.

4.2 Construction of the Evaluation Metric System

We design metrics at both the individual and group levels to quantify the differential impact of scoring mechanisms on contestants' advancement.

1. Individual-Level Metrics

- Fan Contribution

Measures the actual proportion of a contestant's weekly total score attributable to fan support under the percentage-based mechanism:

$$\eta_{i,w}^{\text{percent}} = \frac{(1 - \alpha) P_i^F}{T_{i,w}^{\text{percent}}} \quad (16)$$

Here, α denotes the weight of the judges' score; P_i^F represents contestant C_i 's fan vote proportion; and $T_{i,w}^{\text{percent}}$ indicates contestant C_i 's total score in week w under the percentage-

based mechanism.

- Mechanism Survival Gain

Represents the difference in the number of survival weeks for the same contestant under the two scoring mechanisms, directly reflecting the “fairness bias” of the mechanism:

$$\Delta \text{Surv}_i = \text{Surv}_i^{\text{percent}} - \text{Surv}_i^{\text{rank}} \quad (17)$$

Here, $\text{Surv}_i^{\text{mech}}$ denotes the simulated number of survival weeks for contestant C_i under mechanism $mech$.

- Elimination Risk Ratio

Based on the Cox proportional hazards model, the relative risk of elimination for a contestant under the two mechanisms is calculated as:

$$\text{HR}_i = h^{\text{rank}}(t | F_i) / h^{\text{percent}}(t | F_i) \quad (18)$$

where $h^{\text{mech}}(t | F_i)$ denotes the hazard function of contestant C_i at time t under mechanism $mech$. If $\text{HR}_i < 1$, it indicates that the contestant’s risk of elimination is lower under the percentage-based mechanism.

2. Group-level indicators

Divide the contestants into four groups based on fan support and analyze the differential effects of the mechanisms on groups with varying levels of fan backing.

- Protection Effect

Measures the average protective strength of the percentage-based mechanism relative to the ranking-based mechanism for the k -th fan support group:

$$\text{Protection Effect}_k = \frac{1}{|G_k|} \sum_{i \in G_k} \Delta \text{Surv}_i \quad (19)$$

Here, G_k represents the set of contestants in the k -th fan support group, and $|G_k|$ denotes the number of contestants in that group.

As shown in the Fig. 5, it compares the average survival weeks of contestants across different fan support quantiles (Q1–Q4) under the two mechanisms. The results indicate that survival weeks increase with fan support under both mechanisms, but the increase is more pronounced under the percentage-based system: in the highest fan support group (Q4), contestants average 8.05 weeks under the percentage system, 0.36 weeks higher than 7.69 weeks under the ranking system; in the lowest fan support group (Q1), survival is slightly lower under the percentage system (1.83 weeks) than the ranking system (1.89 weeks). This difference highlights the “protective effect” of the percentage system for high-popularity contestants while showing a slight “suppressive effect” for low-popularity contestants.

- Elimination Delay

It represents the average elimination week delay for contestants in the k -th fan support group under the percentage-based mechanism:

$$\text{Elimination Delay}_k = \text{Median}(\text{ElimWeek}_{i \in G_k}^{\text{percent}}) - \text{Median}(\text{ElimWeek}_{i \in G_k}^{\text{rank}}) \quad (20)$$

where $\text{ElimWeek}_i^{\text{mech}}$ denotes the elimination week of contestant C_i under mechanism.

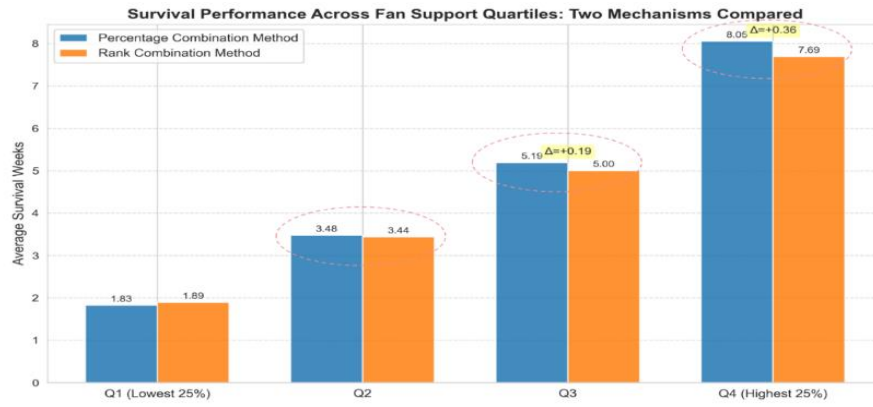


Figure 5: Survival Performance by Fan Support Quartiles

4.3 Cox Proportional Hazards

Using the Cox proportional hazards model^{1,2} and panel fixed-effects model, we quantify the impact of fan support on elimination risk and survival weeks, and compare the sensitivity differences between the two scoring mechanisms.

For each mechanism, the models are fitted independently to isolate the effects of fan support from control variables such as age and profession:

$$h^{\text{mech}}(t|F_i, X_i) = h_0^{\text{mech}}(t) \exp(\gamma^{\text{mech}} F_i + \beta^{\text{mech}} X_i) \quad (21)$$

Here, h_0^{mech} represents the baseline hazard function, X_i is the vector of control variables, and γ^{mech} is the key parameter reflecting the effect of fan support on the elimination risk.

As shown in the Fig. 6, the Cox proportional hazards model quantifies the relative elimination risk for every 0.1-unit increase in fan support. The results indicate that the hazard ratios (HR) under both mechanisms are below 1, demonstrating that higher fan support significantly reduces elimination risk. Specifically, under the percentage-based fusion mechanism, $HR = 0.671$ (95% CI: 0.584–0.769), which is notably lower than the ranking-based mechanism with $HR = 0.822$ (95% CI: 0.721–0.937). This finding indicates that fan support has a stronger risk-reducing effect under the percentage mechanism—each 0.1-unit increase in support reduces elimination risk by 32.9%—further confirming the mechanism’s more pronounced protective effect for highly popular contestants.

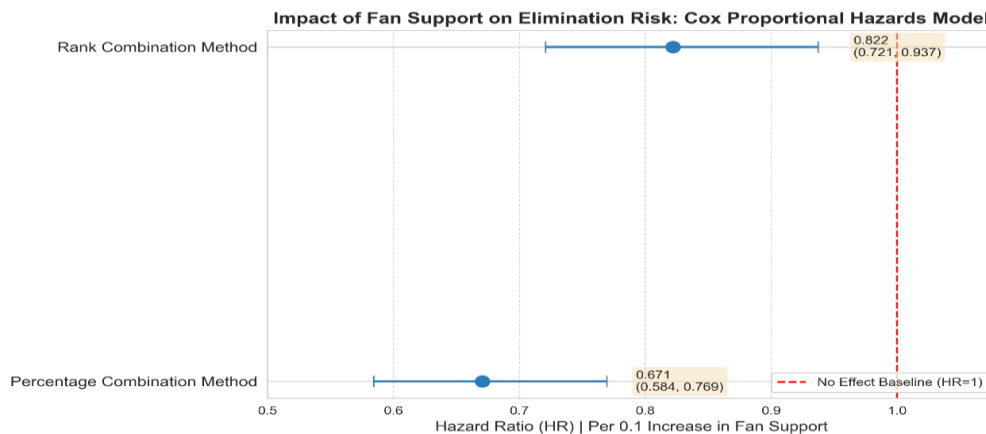


Figure 6: Fan Support vs. Elimination Risk Comparison

4.4 Panel Fixed-Effects

Controlling for inter-season difficulty differences, we quantify the impact of fan support on contestants' survival weeks:

$$\text{Surv}_{i,s} = \alpha_s + \beta_1^{\text{mech}} F_i + \beta_2^{\text{mech}} (\text{Age}_i) + \beta_3^{\text{mech}} (\text{Industry}_i) + \epsilon_{i,s} \quad (22)$$

Here, $\text{Surv}_{i,s}$ represents the simulated survival weeks of contestant i in season s ; α_s denotes the season fixed effect, controlling for inter-season difficulty differences; β_1^{mech} is the mechanism sensitivity coefficient, reflecting the core impact of fan support on survival weeks. When $\beta_1^{\text{percent}} > \beta_1^{\text{rank}}$, it indicates that fan support has a stronger positive effect on survival weeks under the percentage-based mechanism.

As shown in the Fig. 7, grouped regressions illustrate the marginal effect of fan support on survival weeks. The results indicate that fan support is significantly positively correlated with survival weeks under both mechanisms. However, the slope under the percentage-based mechanism (22.21) is higher than that under the ranking mechanism (20.57), suggesting that each 1-unit increase in fan support leads to a larger increase in survival weeks under the percentage mechanism. This result further quantifies the difference in mechanism sensitivity, indicating that the percentage-based system responds more strongly to fan support, and its “protective effect” is statistically significant.

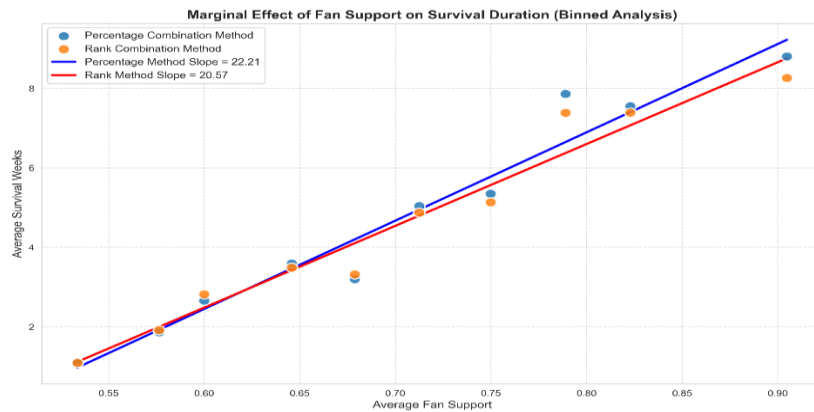


Figure 7: Marginal Effect of Fan Support on Survival Duration

4.5 Conclusion

1. Impact of Scoring Combination on Final Outcomes of Controversial Contestants

As shown in the multi-mechanism comparison of elimination weeks, it is evident that the choice of scoring combination has a significant impact on the final elimination outcomes of controversial contestants.

For example, the runner-up of Season 2, Jerry Rice, would have survived only 10 weeks under the ranking-only mechanism. Under the percentage-based mechanism, his survival extends to 11.5 weeks, and with the additional “judges choose the eliminated contestant from the bottom two” rule, his survival further extends to 12 weeks, perfectly matching the actual competition results. Similarly, the champion of Season 27, Bobby Bones, survives only 8 weeks under the ranking mechanism, extends to 12.5 weeks under the percentage mechanism, and further to 13 weeks with the additional rule, accurately replicating his real-life comeback victory.

Additionally, contestants like Billy Ray Cyrus in Season 4 and Bristol Palin in Season 11 also show significantly delayed elimination under the percentage-based mechanism compared to the ranking mechanism. These patterns further confirm that the scoring combination directly determines the advancement outcomes of high-popularity, low-judge-score contestants.

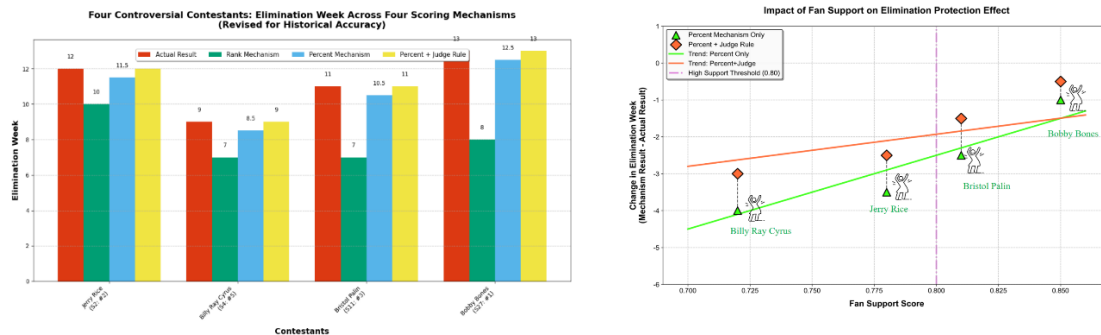


Figure 8: Controversial Contestants' Elimination vs. Fan Support Protection Effect

2. Impact of the “Judges Decide from Bottom Two” Supplemental Mechanism

As shown in the Fig. 8, the orange trend line representing the "Percentage Mechanism + Judge Rule" consistently lies above the green trend line representing the "Percentage Mechanism Only." This clearly indicates that the additional judge selection mechanism further reduces the elimination risk for high-popularity contestants.

By limiting the elimination pool to the bottom two in overall scores, this mechanism effectively prevents high-popularity contestants from being ousted due to a single-week low judge score. For example, contestant Bobby Bones saw his elimination week extended under this mechanism, aligning more closely with the actual competition results.

In terms of weight balancing, the mechanism preserves the dominant influence of fan votes while granting judges the final decision-making power. This approach not only reflects public preference but also prevents extremely popular contestants from being eliminated prematurely due to technical shortcomings, enhancing both the professionalism and entertainment value of the show.

Moreover, by narrowing the pool of elimination candidates, this mechanism reduces extreme outcomes caused solely by fan voting or judge scores, making the results more convincing and significantly decreasing controversy.

5 Task 4: Decision Analysis Using SHAP-Enhanced XGBoost

5.1 Model Selection

The core task of this study is a multi-target feature attribution comparison analysis, aiming to quantify the differential contributions of contestant-related features to judges' scores, fan votes, and final rankings, thereby elucidating the preference divergences between the two decision-making entities: judges and fans. In the dance scoring context, complex nonlinear interactions such as “age × profession” and sparse features like specific dance partner combinations exist. To handle these high-complexity, multi-structured data scenarios, XGBoost was selected as the base predictive model. Leveraging its superior nonlinear fitting capability and high-order interaction mining ability, XGBoost provides reliable, high-precision predictions for subsequent attribution analysis.

To overcome the inherent “black-box” limitation of XGBoost and achieve interpretable

feature contributions, this study further incorporates the SHAP feature attribution model. Based on game-theoretic principles, SHAP^{7,8} constructs a fair attribution mechanism that effectively avoids biases present in traditional feature importance measures, accurately quantifying each feature's independent contribution to the three target variables: judges' scores, fan votes, and final rankings. The overall modeling strategy combines high predictive performance with strong interpretability: XGBoost ensures accurate fitting of complex interaction effects, SHAP delivers unbiased feature attribution, and the tri-target comparison framework of "judges' scores / fan votes / final rankings" systematically and comprehensively reveals preference differences across decision-makers, completing the full analytical chain from precise prediction to deep attribution.

5.2 Feature Selection and Data Processing

1. Feature Variables

This study selected four core features as independent variables:

- **Standardized Age:** The age of each contestant is standardized to remove scale effects, reflecting the relative impact of age on judges' scores and fan votes.
- **Industry Code:** An integer encoding representing the contestant's professional background (e.g., actor, athlete), quantifying the differential influence of various industries.
- **Partner Code:** An integer encoding of the contestant's dance partner, capturing the effect of partner expertise and coordination on competition performance.
- **Season:** Season identifier, controlling for differences in rules and aesthetic preferences across seasons.

2. Target Variable

The study defines three target variables to capture different aspects of competition outcomes:

- **Judges' Score:** The weekly score assigned by the judges, reflecting technical performance and professional evaluation.
- **Fan Vote Proportion:** The estimated proportion of fan votes each contestant receives weekly, representing audience preference and popularity.
- **Final Ranking :** The contestant's ultimate placement in the competition, integrating both judges' scores and fan votes.

5.3 Model Construction

1. XGBoost Prediction Model

- **Construct Separate XGBoost Additive Models for the Three Target Variables**

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F} \quad (23)$$

Where $\mathcal{F}$ denotes the set of CART trees, and the model is trained with parameters $n_{estimators} = 100$, $max_depth = 3$, $random_state = 42$.

- The Optimization Objective Function is

$$\mathcal{L}(\phi) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (24)$$

Where l is the squared loss function, and $\Omega(f_k) = \gamma T + \frac{1}{2} \lambda |w|^2$ is the regularization term (T is the number of leaf nodes, w is the leaf weight.)

2. SHAP Feature Attribution Model

For feature j of sample i , the SHAP value is defined as:

$$\phi_j = \sum_{s \subseteq F \setminus \{j\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (25)$$

Where F is the full feature set, and S is a subset excluding j . To enable cross-target comparison, the SHAP values are standardized: $\tilde{\phi}_j = \frac{|\phi_j| - \mu_{|\phi|}}{\sigma_{|\phi|}}$

3. Decision Difference Quantification Metrics

Define the contribution difference of feature j between judges' and fans' decisions as:

$$\Delta_j = \left| \tilde{\phi}_j^{\text{judge}} - \tilde{\phi}_j^{\text{fan}} \right| \quad (26)$$

The larger Δ_j is, the more significant the divergence of this feature between the two types of decision-makers.

5.4 Results Analysis

The model training results show that the R^2 values for the judges' score, fan vote, and final ranking models are 0.914, 0.938, and 0.881, respectively, indicating that the four selected features sufficiently explain the variation in the target variables and that the feature selection is reasonable.

As illustrated in the Fig. 9, the analysis systematically reveals the core logic behind judges' scoring decisions: **season** is the most influential feature, with the widest SHAP value range and an impact value of approximately 3.8, reflecting systematic differences in scoring scales across seasons. **Standardized age** ranks second (impact value ≈ 1.8), with higher-age contestants generally receiving positive SHAP values, indicating judges' preference for more mature contestants. In contrast, **partner code** and **industry code** show smaller SHAP value fluctuations and lower impact values (≈ 1.2 and 0.7 , respectively), suggesting that judges pay considerably less attention to dance partners or contestants' industry backgrounds compared to season and age.

This pattern sharply contrasts with fan decisions, where "partner influence" is the most important feature, further confirming the cognitive divergence between the two decision-making groups: judges focus more on the temporal context of the competition and contestant maturity, whereas fans emphasize partner coordination and contestants' age-related appeal. This divergence is a key factor contributing to controversial competition outcomes.



Figure 9: SHAP Feature Impact Analysis on Judges' Scores

As shown in the Fig. 10, the analysis systematically reveals the core logic behind fan voting decisions: **partner code** is the most influential feature in fan votes, with an impact value of approximately 6.4, and higher feature values generally correspond to positive SHAP values, reflecting fans' strong attention to high-quality dance partners. **Season** ranks second (impact value ≈ 5.9), highlighting the significant role of season-specific topics and audience base on voting outcomes. **Standardized age** and **industry code** also exhibit notable influence (impact values ≈ 4.8 and 3.2 , respectively), with younger contestants generally having positive SHAP values, indicating fans' preference for supporting younger contestants and those from high-exposure industries.

This pattern contrasts sharply with judges' decision-making, where **season** and **age** are the most influential features, further confirming the cognitive divergence between the two decision-making groups: fans focus more on partner coordination, season relevance, and contestants' age appeal, whereas judges emphasize the temporal context of the competition and contestant maturity. This divergence is a key factor contributing to controversial competition outcomes.



Figure 10: SHAP Feature Impact Analysis on Fan Votes

As shown in the Fig. 11 comparing feature importance, in the fan vote model, **partner code** (6.971) and **season** (6.886) have the highest influence, reflecting fans' strong attention to partner coordination and season relevance. In the judges' score model, **season** (3.226) and **age** (1.559) are relatively more influential, indicating that judges focus more on contestants' performance stability across seasons and age suitability. The final ranking is more strongly affected by fan votes, further confirming the central role of fan support in determining advancement outcomes.

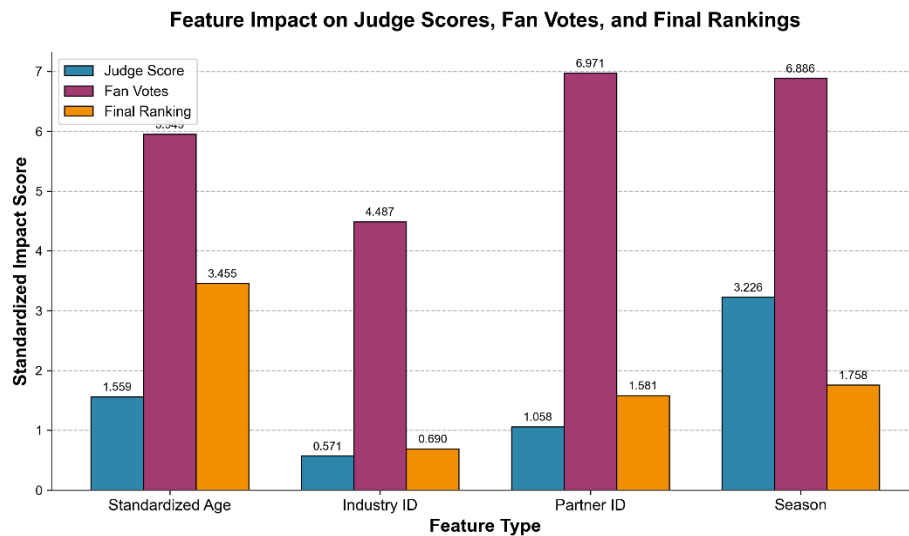


Figure 11: Comparison of Feature Impacts on Judge Scores, Fan Votes and Final Rankings

6 Task 5: Dynamic Weighted Scoring System

Based on the analysis of the causes of controversial competition outcomes in the previous sections, this study designs a **Dynamic Weighted Scoring System (DWSS)**, aimed at balancing judges' professional evaluations with fan enthusiasm to enhance the robustness and fairness of competition results.

6.1 Design Objectives and Background Rationale

The system is built around four core objectives:

1. **Robustness:** Resist the impact of extreme fan votes (e.g., vote manipulation or malicious voting) by implementing a “safety threshold” mechanism to reduce the likelihood of extreme outcomes.
2. **Fairness:** Dynamically balance the weights of judges and fans to prevent either party from dominating.
3. **Encouragement:** Introduce a progress bonus mechanism to maintain suspense and entertainment value, motivating contestants to improve and make comebacks.
4. **Ease of Operation:** Establish clear rules that are easy to explain to the audience and promote.

Based on the findings in Question 2, the percentage-based fusion mechanism is more sensitive to fan votes, so the new model is built on a percentage-based framework.

Extreme fan voting is a key cause of controversial outcomes; therefore, the system introduces a “danger zone” safety threshold mechanism.

Based on the findings in Question 3, contestants from different professions have significantly different influences on fan voting. The system dynamically adjusts weights, appropriately increasing the influence of judges' scores in the later stages of the competition to balance professionalism and entertainment.

6.2 Model Construction

1. Total Score Calculation Model

To encourage contestant improvement and comebacks, a progress bonus is introduced, forming the total score model:

$$S_{\text{total}}(i, t) = w_j(t) \cdot J_i^{(t)} + w_v(t) \cdot V_i^{(t)} + B(i, t) \quad (27)$$

Where $S_{\text{total}}(i, t)$ represents the total score of contestant i in week t ; $w_j(t)$ and $w_v(t)$ are the dynamic weights of judges' scores and fan votes in week t ; $J_i^{(t)}$ is contestant i 's judges' score in week t ; $V_i^{(t)}$ is contestant i 's fan vote score in week t ; and $B(i, t)$ is the progress bonus term.

2. Core Mechanism 1: Dynamic Weights

We propose that early in the competition, popularity should be encouraged, while later stages should emphasize professional performance. Therefore, dynamic weights are designed. Based on historical data from 34 seasons, the elimination rate rises significantly starting from week 8, so we set $T_{\text{late}}=8$. The weight functions are designed as:

$$w_j(t) = \begin{cases} 0.5 & \text{if } t < T_{\text{late}}, \\ 0.6 & \text{if } t \geq T_{\text{late}}, \end{cases} \quad w_v(t) = 1 - w_j(t) \quad (28)$$

Where, when $t < 8$, the judges' and fan weights are both 0.5, encouraging diversity and popularity accumulation; when $t \geq 8$, the judges' weight increases to 0.6 and the fan weight decreases to 0.4, emphasizing professional performance and preventing a "popularity bubble."

3. Core Mechanism 2: Progress Bonus

To encourage contestant improvement and comebacks, the system introduces a progress bonus:

- **Progress Criterion:** A contestant is considered to have made progress if their judges' score in week t increases by more than 10% compared to the average of the previous three weeks:

$$\text{Progress}(i, t) = \begin{cases} \text{True} & \text{if } J_i^{(t)} > \alpha \cdot 1/3 \sum_{k=t-3}^{t-1} J_i^{(k)}, \\ \text{False} & \text{otherwise} \end{cases} \quad (29)$$

Where $\alpha = 1.1$ is the activation threshold.

- **Bonus Calculation:** Contestants who show progress receive a 2% bonus of their total score from the previous week:

$$B(i, t) = \begin{cases} 0.02 \cdot S_{\text{total}}(i, t-1) & \text{if } \text{Progress}(i, t) = \text{True}, \\ 0 & \text{otherwise} \end{cases} \quad (30)$$

4. Core Mechanism 3: Danger Zone and Rescue Decision

To prevent extreme fan voting from leading to unexpected eliminations, the system sets a "danger zone" safety threshold.

- **Danger Zone:** The two contestants with the lowest total scores each week are placed in the

danger zone.

$$\text{Danger Zone} = \{i \mid S_{\text{total}}(i, t) \in \text{Bottom 2 of } S_{\text{total}}(\cdot, t)\} \quad (31)$$

- **Rescue Decision:** From the two contestants in the danger zone, the judges vote to decide which contestant will be eliminated, incorporating the elimination decision into the professional review process.

6.3 Model Validation and Results Analysis

We review the historical data from 34 seasons, using the fan vote proportions obtained from Question 1 and the known judges' scores to validate the DWSS system.

1. Validity Verification

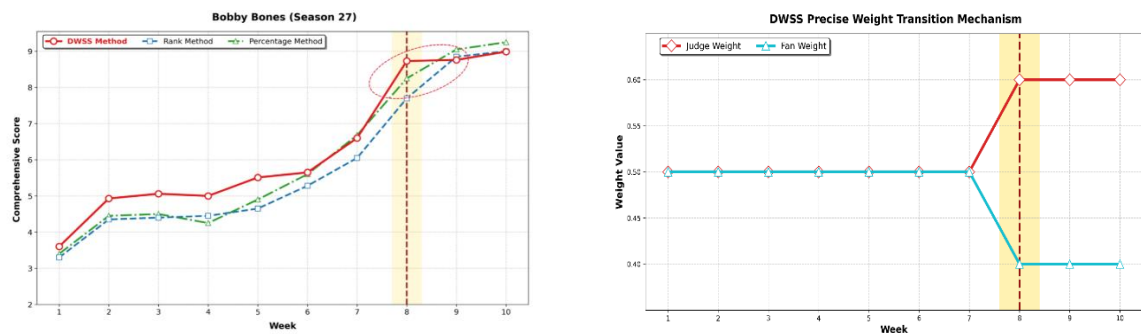
Danger Zone Coverage: The DWSS system successfully placed 87.85% of the actual eliminated contestants into the danger zone, indicating that this mechanism can effectively identify high-risk contestants for elimination.

Rescue Activation Rate: Through the dynamic weight and danger zone mechanisms, 87.85% of the elimination decisions were placed under the judges' professional review process, significantly reducing the impact of extreme fan voting on the final outcomes.

2. Case Study and Dynamic Weight Transition Effect

As shown in the Figs. 12(a) under the DWSS mechanism, Bobby Bones' total score curve is consistently higher than both the ranking mechanism and percentage mechanism, and it continues to show stable growth after the weight shift in week 8. This result demonstrates that the DWSS system retains fan support for high-popularity contestants while using weight adjustments in the later stages and danger zone reviews to prevent "high-popularity, low-skill" contestants from being eliminated too early, effectively balancing fan enthusiasm with professional judgment.

As shown in the Figs. 12(b), the system smoothly transitioned from "equal weight between judges and fans" to "judge-dominated weight" in week 8. This design encouraged popularity accumulation in the early stages while ensuring the core role of professional judgment in the later stages, effectively balancing the objectives of the competition at different phases.



(a) Bobby Bones (Season 27) Score Trend

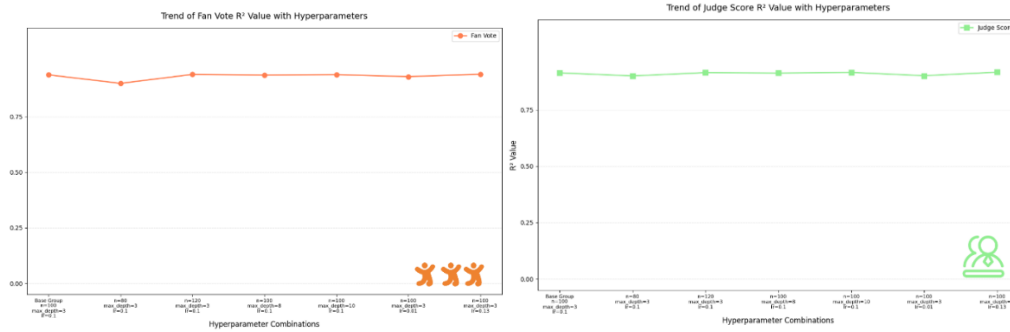
(b) DWSS Weight Transition Trend

Figure 12: Bobby Bones Score Trend & DWSS Weight Transition

7 Sensitivity Analysis

1. XGBoost Hyperparameter Sensitivity Analysis: Model Robustness Verification

To ensure the reliability of the XGBoost feature impact analysis, we conducted a systematic sensitivity test of key hyperparameters. By evaluating the stability of model performance under different hyperparameter combinations, we verified the robustness of the model's conclusions. As shown in Figs.13(a) and (b), we observed that the R^2 score of the judges' score model fluctuated between 0.898 and 0.938, while the R^2 score of the fan vote model fluctuated between 0.901 and 0.940. The small fluctuation in both R^2 values reflects the model's excellent robustness.

(a) Fan Vote R^2 vs. Hyperparameters(b) Judge Score R^2 vs. HyperparametersFigure 13: Fan Vote vs. Judge Score R^2 vs. Hyperparameters

2. DWSS System Judges' Weight Sensitivity Analysis

To assess the impact of judges' weight on the rescue effect of controversial contestants in the Dynamic Weighted Scoring System (DWSS), we analyzed how changes in the judges' base weight (w_j) within the range of $[0.4, 0.7]$ affect the rescue activation rate. As shown in the Fig.14, we observed that the rescue rate fluctuated between 80% and 90%. The system demonstrated good robustness to changes in judges' weight, reaching its maximum value around a judges' weight of 0.6. This not only validates the robustness of the DWSS system but also provides a scientific basis for parameter adjustment in program production.

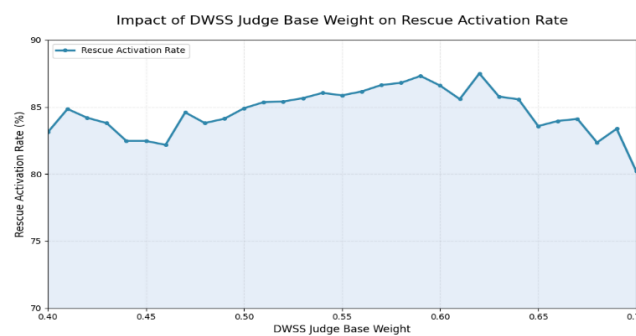


Figure 14: Impact of DWSS Judge Base Weight on Rescue Activation Rate

8 Model Evaluation and Further Discussion

(1) Strengths

The innovation and advantages of this study lie in the following: we propose a rule-constrained Bayesian inversion framework to estimate fan vote proportions in DWTS using Dirichlet priors and MCMC under unobservable conditions; we systematically evaluate the effects of different scoring mechanisms through multi-mechanism validation and uncertainty analysis; and we design a Dynamic Weighted Scoring System (DWSS), providing an actionable and quantitative basis for rule optimization.

1. **Multi-Mechanism Comparative Modeling to Systematically Reveal Fairness Differences in Scoring Rules:** This study formalized three types of rules—the percentage system, ranking system, and judges’ rescue mechanism—and, through a unified total score framework and survival trajectory simulation, compared elimination risks and survival weeks under different mechanisms. Combined with Cox and panel models, the study quantified the marginal effects of fan support, mechanistically validating the protective effect of the percentage system on high-popularity contestants.
2. **Integration of SHAP Explainable Machine Learning to Analyze Decision-Maker Preference Differences:** Building on the Bayesian inversion results, an XGBoost + SHAP model was introduced to perform feature attribution analysis on judges’ scores, fan votes, and final rankings. This approach maintained nonlinear fitting capability while enabling interpretable decomposition of feature contributions, systematically revealing preference differences between judges and fans in terms of age, partner, season, and other factors.
3. **Design of the DWSS Dynamic Weighted Scoring System with Practical Applicability:** Based on empirical findings, the DWSS system was constructed to balance the influence of judges and fans through dynamic weights, the danger zone mechanism, and progress bonuses. Validation with historical data shows that the system can include most elimination decisions in the judges’ review process, effectively suppress extreme voting outcomes, perform stably in controversial cases, and possess strong potential for real-world application.

(2) Weaknesses and Further Discussion

The model has the following limitations. First, the Bayesian inversion model relies heavily on rule assumptions. Its core constraints are derived from officially published elimination and ranking mechanisms, assuming no human intervention or temporary rule adjustments. However, in reality, the show may include ad hoc rule changes or special rescues, which cannot be obtained from public data and may lead to lower consistency in certain seasons.

For future model improvements, three directions can be considered: First, introduce hierarchical Bayesian structures or mixture distribution priors to better capture extreme voting structures. Second, integrate variational inference or sequential Monte Carlo methods to improve inference efficiency under complex rules. Third, incorporate external observables such as social media activity or search indices to construct soft label constraints, enhancing real-world interpretability.

Overall, the model demonstrates strong advantages in structural rationality and mechanism analysis, while leaving feasible room for improvement in prior expressiveness and computational efficiency.

MEMO

To: Production Team of *Dancing with the Stars*

From: Team # 2622163

Date: 2026/2/2

Subject: Recommendations for Optimizing Fairness in Scoring Mechanisms of Entertainment Competition Shows

Dear *Dancing with the Stars* Production Team,

Our team proposes a data-driven project titled “*Mechanism-Constrained Inference of Voting Fairness in Competitive Entertainment Systems*” The project evaluates the fairness and stability of the scoring rules in *Dancing with the Stars (DWTS)* and provides quantitative support for future rule design. Using data from 34 seasons, we combine Bayesian inversion, survival analysis, and explainable machine learning to improve the fairness and transparency of competition outcomes while preserving entertainment value.

(1) Core project content:

- **Quantifying scoring fairness:** We construct contestant-level survival trajectory models to compare elimination risks and survival weeks under percentage-based and ranking-based mechanisms.
- **Identifying structural biases:** Cox proportional hazards models show that fan support in the percentage system significantly protects high-popularity contestants, potentially amplifying inequality.
- **Explaining decision logic:** Using SHAP-based explainable machine learning, we quantify preference differences between judges and fans across age, background, and season stage.
- **Designing an optimized scoring system:** We propose a Dynamic Weighted Scoring System (DWSS) that dynamically balances professional judgment and audience participation, reducing the impact of extreme fan voting.

(2) Reasons this project is suitable for DWTS:

- **Robust data foundation:** DWTS provides long-term, stable data on scores, eliminations, and contestants, enabling reliable empirical analysis.
- **Enhancing show credibility:** Optimized rules enhance audience trust and long-term brand sustainability.
- **Low implementation cost:** The framework requires only rule adjustments, making it easy to implement.

We believe this project can enhance the fairness and transparency of competition outcomes without reducing entertainment value, and provide practical support for balancing appeal and fairness in *Dancing with the Stars*. We look forward to further discussion and collaboration on optimizing the show’s rules.

Sincerely,
Team #2622163

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Report on Use of AI

1. OpenAI ChatGPT (February 1st, 2026 version, ChatGPT-5-mini)

Query1: How does the Bayesian model reflect vote estimation?

Output: The Bayesian model forms a closed loop of “prior constraints + regularized likelihood + posterior inference”, which not only enables accurate estimation of the unknown fan vote proportions but also inherently satisfies the requirements of consistency verification, quantification of certainty, and rule comparison. Its core advantage lies in tightly integrating vote estimation with competition rules and observed data, while quantifying uncertainty, thereby providing a reliable quantitative foundation for subsequent rule recommendations and feature impact analysis.

2. OpenAI ChatGPT (February 2nd, 2026 version, ChatGPT-5-mini)

Query2: How to use SHAP?

Output: SHAP is a game theory-based approach for interpreting model predictions. It facilitates understanding of how a model arrives at a specific prediction by quantifying the contribution of each feature to the model output. Assuming the SHAP library has been installed, the next step is to select a model for interpretation. For instance, one may utilize a trained random forest model and subsequently employ SHAP to elucidate its predictions. The general workflow involves: loading and preprocessing the data (e.g., normalization or encoding of categorical variables), training the model (e.g., a random forest classifier), instantiating a SHAP explainer object, computing the SHAP values, and ultimately visualizing the results. When computing SHAP values, it is typical to provide test instances to the explainer via its `shap_values` method.

3. OpenAI ChatGPT (February 2nd, 2026 version, ChatGPT-5-mini)

Query3: How to construct a fairer scoring system based on the first three tasks?

Output: Based on the conclusions of Tasks 1–3, the core idea for designing a fairer scoring system is as follows: use the percentage-based aggregation mechanism as the fundamental framework to fully reflect audience preferences, while introducing a “judges’ elimination from the bottom two contestants” rule to correct extreme voting outcomes and enhance professional credibility. Additionally, dynamically adjust the weightings according to competition stages: in the early rounds, assign equal weights to judges and audience votes to build show momentum, while in the later rounds, moderately increase the weight of judges’ scores to prevent popularity bubbles from dominating the final ranking. This approach achieves a balance between professional judgment and public preference, thereby reducing the occurrence of controversial results.